

**DESCRIPTION**

APZ 4420 is a pressure transmitter with a WECO® 2" (Hammer Union) connection. The monolithic process connection prevents the transmitter from changing its characteristics when mounted in harsh environments. Thin-film piezoresistive sensor ensures low drift and the best long-term stability. The pressure transmitter is particularly resistant to strong vibrations, shocks and pressure surges. It is reliable in cementing and wellbore tightening applications.

SPECIFICATIONS

Pressure ranges: 0...5000 psi to 0...15000 psi

Basic accuracy: $\pm 0.5\%$

Outputs: 4...20 mA

Sensor: thin-film piezoresistive

Pressure port: WECO® 2" (1502)

Media temperature: $-40...+125\text{ }^{\circ}\text{C}$

Ambient temperature: $-40...+125\text{ }^{\circ}\text{C}$

Optional: Ex ia

APPLICATIONS

Onshore and offshore applications in the petrochemical industry

Fracking equipment

Acid Treatment Equipment

Drilling mud/slurry pressure measurements

WECO® is a registered trademark of the TechnipFMC.

ORDERING CODE

MEASURING RANGES

Pressure range, psi	Overpressure, psi	Burst pressure, psi
Gauge		
0...5000	7500	10000
0...6000	9000	12000
0...10000	15000	20000
0...15000	22500	30000

PERFORMANCE

Accuracy, % of span*	≤ ±0.5 (±0.25% of span BFSL)
Temperature effect (% of span / 10 °C)	≤ ±0.2
Compensated range	-20...+70 °C
Power supply effect	≤ ±0.05% of span / 10 V
Load resistance effect	≤ ±0.05% of span / kΩ
Long-term stability	≤ ±0.2% of span / year
Response time (10...90%)	≤ 1 ms

*Accuracy includes non-linearity, hysteresis and non-repeatability (IEC 60770). Calibration certificate available optionally

OPERATING CONDITIONS

Medium temperature	-40...+125 °C (Ex: -40...+85 °C)
Ambient temperature	-40...+125 °C (Ex: -40...+85 °C)
Storage temperature	-55...+125 °C
Approval	0Ex ia IIC T4 Ga X
Vibration resistance	10 g RMS, 25–2000 Hz (DIN EN 60068-2-6)
Shock resistance	50 g / 1 ms (DIN EN 60068-2-27)
Service life	> 100 x 10 ⁶

MECHANICAL SPECIFICATIONS

Pressure port material	stainless steel 1.4548 (17-4PH)		
Housing material	stainless steel 304 (1.4301)		
Diaphragm	stainless steel 1.4548 (17-4PH)		
Wetted parts	Diaphragm, pressure port		
Pressure port	WECO® 2" (1502)		
Electrical connection	Ingress protection	Cross section	Cable diameter
MIL-/Bendix (PT02-E10-6P-023), 6 pin	IP67	0.75 mm²	6...8 mm

ELECTRICAL SPECIFICATIONS

	Output signal	Power supply, U _s	Load resistance	Power consumption
	4...20 mA / 2-wire	12...32 V	= (U _s – 12 V) / 0.02 A Ω	≤ 26 mA
Ex ia	4...20 mA / 2-wire	14...28 V	=(U _s – 14 V) / 0.02 A Ω	≤ 26 mA

Safe values for intrinsically safe design Ex ia:

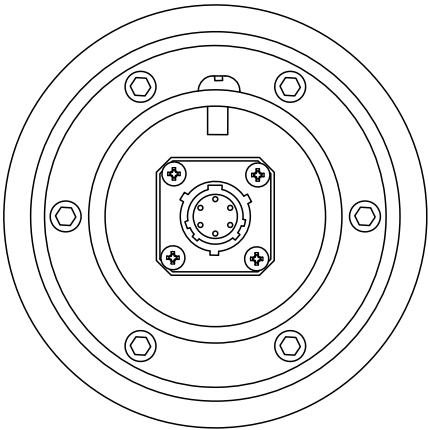
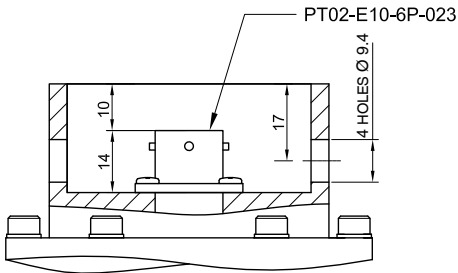
Parameter	Value
Maximum voltage, U _i	28 V
Maximum current, I _i	93 mA
Maximum power, P _i	660 mW
Maximum internal inductance, L _i	10 μH
Maximum internal capacitance, C _i	25 nF
Maximum DC voltage or effective value of AC voltage, U _m	28 V

ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

	Circuits	MIL-/Bendix (type PT02-E10-6P-023) connector, 6 pin.
2-wire	power +	A
	power -	B
	shield	D

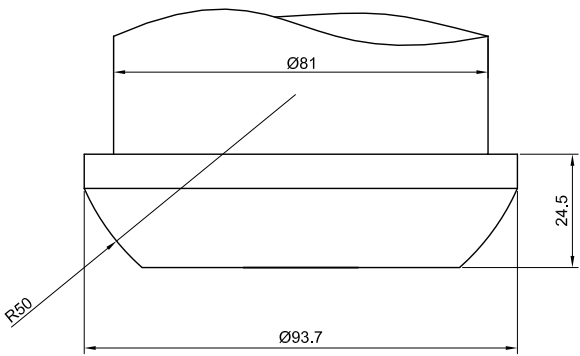
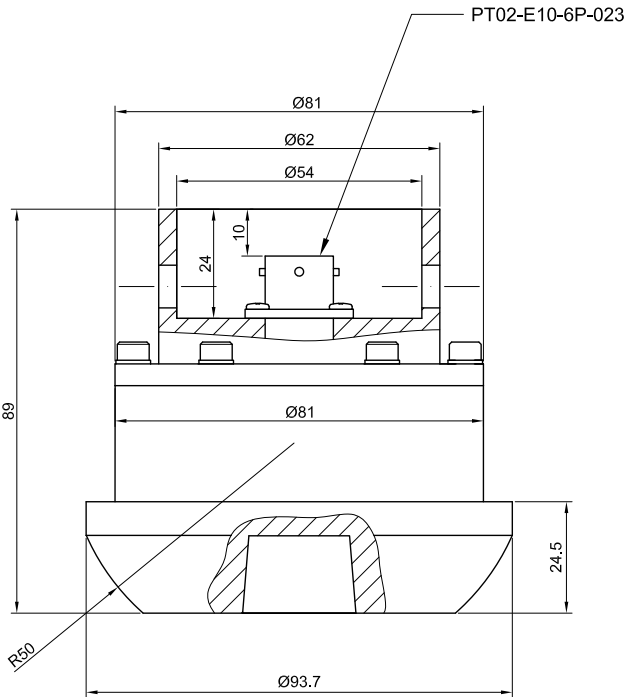
ELECTRICAL CONNECTIONS, DIMENSIONS (mm)

PT02-E10-6P-023



DIMENSIONS (mm)

PRESSURE PORTS, DIMENSIONS (mm)



ORDERING CODE

	APZ 4420	-X	-X	-XXXX	-X	-XX	-X	-XXX	-XX
MEASUREMENT TYPE									
	Gauge	G							
UNIT OF MEASUREMENT									
		psi	N						
	Other (specify when ordering)		X						
UPPER RANGE LIMIT (URL), psi									
		0...5000	5003						
		0...6000	6003						
		0...10000	1004						
		0...15000	1504						
	Other (specify when ordering)		XXXX						
ACCURACY									
		±0.5 % (±0.25% BFSL)	D						
	±0.5 % (±0.25% BFSL) with calibration certificate		M						
	Other (specify when ordering)		X						
ELECTRICAL CONNECTION									
	MIL-/Bendix (PT02-E10-6P-023), 6 pin / IP 67		MB						
	Other (specify when ordering)		XX						
OUTPUT SIGNAL									
		4...20 mA / 2-wire / 12...32 V	A						
		4...20 mA / 2-wire / 14...28 V + Ex ia	Q						
	Other (specify when ordering)		X						
PRESSURE PORT									
		WECO® 2" (1502)	W15						
	Other (specify when ordering)		XXX						
VERSION									
			Standard	00					
		Other (specify when ordering)		XX					

Example: APZ 4420-G-N-5003-D-MB-A-W15-00

ACCESSORIES

				
PZ 1024 Power supply unit				